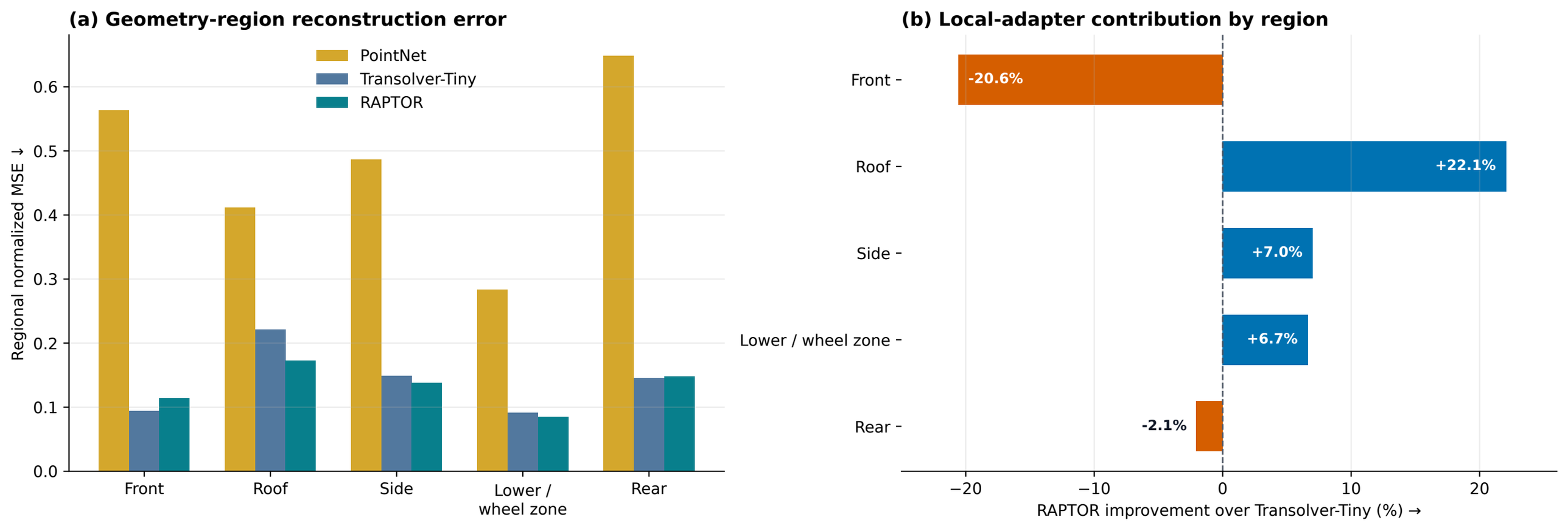
(a) Geometry-region reconstruction error
PointNet
Transolver-Tiny
RAPTOR
Regional normalized MSE ↓
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Front
Roof
Side
Lower / wheel zone
Rear
(b) Local-adapter contribution by region
Front
Roof
Side
Lower / wheel zone
Rear
-20.6%
+22.1%
+7.0%
+6.7%
-2.1%
−20
−10
0
10
20
RAPTOR improvement over Transolver-Tiny (%) →